

DTSA 5506

Data Mining Project

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Machine Learning Engineer (53.6 / 100)	48.2 / 100
SPECIALIZATION Data Mining Foundations and Practice		

48

CURRENT AVG SCORE
across 6 job roles

88

POTENTIAL SCORE
with recommended additions

+47

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course offers step-by-step guidance and hands-on experience of designing and implementing a real-world data mining project, including problem formulation, literature survey, proposed work, evaluation, discussion and future work.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	53.6	93.1	+39.5	HIGH
Data Scientist	53.2	94.0	+40.7	HIGH
Applied Scientist	52.7	100.0	+47.3	HIGH
AI Engineer	50.1	92.7	+42.6	HIGH
ML Engineer	41.4	74.8	+33.4	HIGH
MLOps Engineer	38.3	76.3	+38.0	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

LLMs & Generative AI

2 of 6 roles

storage

Deep Learning

1 of 6 roles

cnn

Feature Engineering

1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

MEDIUM

Consider LLMs & Generative AI content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.

2

LOW

Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: cnn.

3

LOW

Optional: Feature Engineering — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

53.6 → 93.1 (+39.5 pts)

web

sets

learning

analytics

tools

exploratory

design

development

Data Scientist

53.2 → 94.0 (+40.7 pts)

sets

analytics

presentation

social

problems

Applied Scientist

52.7 → 100.0 (+47.3 pts)

research

storage

problems

models

technical

learning

industry

design

AI Engineer

50.1 → 92.7 (+42.6 pts)

perception

following

models

learning

technical

teams

problems

storage

ML Engineer

41.4 → 74.8 (+33.4 pts)

cnn

learning

machine

models

world

problems

technical

building

MLOps Engineer

38.3 → 76.3 (+38.0 pts)

detection

text

learning

machine

preprocessing

tools

models

development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.